CAMPUSAI

INTRODUCTION TO QUANTUM TECHNOLOGIES AND APPLICATIONS (IQTA)

1. Introduction

Quantum technologies use principles of quantum physics to develop new tools for computing, communication, sensing and measurement. They work with quantum systems such as atoms, photons and other physical systems.

2. Basic Quantum Concepts

- **Qubit:** The basic unit of quantum information. A qubit can be prepared in a superposition of the basis states |0■ and |1■.
- **Superposition:** A quantum state can be a combination of basis states. Measurement gives an outcome with probabilities determined by the state.
- **Entanglement:** A shared quantum state can create correlations between particles that cannot be described as independent states.
- **Measurement:** Observing a quantum system produces a classical result and generally changes the state.
- **Interference:** Quantum probability amplitudes can combine to increase or reduce the probability of outcomes.

3. Main Quantum Technologies

- **Quantum Computing:** Uses qubits and quantum operations to process information.
- **Quantum Communication:** Uses quantum principles for tasks such as quantum key distribution.
- **Quantum Sensing:** Uses sensitive quantum systems for precise measurements of quantities such as time, magnetic fields or acceleration.

4. Applications

Potential application areas include chemistry and materials research, optimisation research, secure communication, precision navigation, medical and scientific measurement, and the development of new sensors. The usefulness of a quantum method depends on the problem and the maturity of the technology.

5. Challenges and Limitations

Quantum devices can be sensitive to noise and environmental disturbance. Building reliable systems requires error correction, stable hardware, specialised skills and careful control. Quantum computers are not faster than classical computers for every task.

6. Example

Two entangled qubits can show correlated measurement outcomes. Measuring one gives information about the joint state's correlations, but this cannot be used to send a controllable message faster than light.

7. Quick Revision Questions

1. What is a qubit?
2. Explain superposition and entanglement.
3. Name three main areas of quantum technology.
4. Mention two challenges in building quantum computers.

Note: These are introductory sample notes. Refer to your course syllabus and class materials for complete coverage.